Connecting to corporate Wi-Fi (802.1X) and guest Wi-Fi

Article KB0010021 | Owner: Network Support | Subcategory: Wi-Fi | Applies to: Windows 11 laptops (Dell Latitude 5440/7440, Lenovo ThinkPad T14/X1 Carbon, HP EliteBook 840), macOS 14/15 laptops (MacBook Pro/Air), Any device that supports 802.1X authentication using a machine certificate

Summary

This article explains how to connect Acme Global laptops to the corporate Wi-Fi network (SSID?ACME-Corp, 802.1X with machine certificate) and to the guest Wi-Fi network (SSID?ACME-Guest). It covers common connectivity problems, step-by-step self-help actions users can take, and the corresponding troubleshooting workflow for Network Support agents.

Symptoms

- Wi-Fi repeatedly drops or disconnects after a few minutes
- Unable to join ACME-Corp; prompts for username/password repeatedly
- Connected to ACME-Corp but no Internet access (limited or no connectivity)
- Wi-Fi shows connected but bandwidth is very low, especially in meeting rooms
- Guest network fails to acquire IP address or shows limited connectivity

Common causes

- Expired or corrupted machine certificate used for 802.1X authentication
- Outdated or incompatible wireless network adapter driver
- Power-saving settings that turn off the radio to save battery
- Incorrectly cached credentials or profile for ACME-Corp
- Interference or channel congestion in high-density areas
- Guest sponsorship not approved or expired sponsor token

Overview

Acme Global provides two wireless networks for employees and visitors. The corporate network, ACME-Corp, uses WPA2-Enterprise with 802.1X authentication and relies on a machine certificate stored in the device's trusted root store. The guest network, ACME-Guest, is an open SSID that requires a sponsor-generated voucher or temporary credentials supplied via the guest portal. Both networks are broadcast from the same campus access points managed by the Network Support team.

Prerequisites

Before attempting any troubleshooting, verify that the device meets the following baseline requirements:

- The laptop is joined to the Acme Global Azure AD / on-premises AD domain.
- The machine certificate for 802.1X is present and valid (issued by the internal CA, not expired).
- Wireless adapter drivers are up to date (Windows Update or vendor utility).
- The device is not in airplane mode and the wireless radio is enabled.
- For guest access, you have a valid sponsor voucher or have completed the self-service guest registration at https://guest.wifi.acme.example.

Self-Help Procedure (User)

Follow these steps to resolve most corporate Wi-Fi connectivity issues. If the problem persists after completing the procedure, contact the IT Service Desk via the portal, phone extension 4357, or the Teams "IT Help" channel.

1. 1. Forget the corporate network
   - Windows: Settings -> Network & Internet -> Wi-Fi -> Manage known networks -> Select ACME-Corp -> Forget.
   - macOS: System Settings -> Network -> Wi-Fi -> Advanced... -> Select ACME-Corp -> - (minus) to remove.
2. Turn off wireless power saving
   - Windows: Device Manager -> Network adapters -> Your wireless adapter -> Properties -> Power Management -> Uncheck "Allow the computer to turn off this device to save power".
   - macOS: System Settings -> Battery -> Power Adapter -> Disable "Put hard disks to sleep when possible" (ensures

radio stays active).
3. Renew the machine certificate (if expired)
- Open Certificates (local computer) -> Personal -> Certificates.
- Locate the certificate issued by "Acme Global Internal CA". If the "Valid to" date is past today, request a renewal from the Network Support team via the self-service portal (Network -> Certificate Renewal) or ask your local IT liaison.
4. Re-join ACME-Corp
- Click the Wi-Fi icon, select ACME-Corp, and click Connect.
- The device should automatically use the machine certificate; no username/password prompt should appear.
5. Test connectivity
- Open a browser and navigate to https://it.acme.example.
- If the page loads, you have Internet access.
- Run a quick ping test (e.g., ping 8.8.8.8) to confirm latency <50?ms.
6. (Optional) Update wireless driver
- Windows: Settings -> Windows Update -> Advanced options -> Optional updates -> Driver updates -> Install any wireless adapter update.
- macOS: Software Update will include driver updates; install any available updates.

If after these steps you still experience drops, no Internet, or repeated credential prompts, proceed to the Service Desk section.

Service Desk Procedure (Network Support)

When a user escalates a Wi-Fi ticket, follow this structured workflow to diagnose and resolve the issue efficiently. Document each step in the ticket.

1. 1. Verify user identity and device
- Confirm the laptop hostname, asset tag, and that it is domain-joined.
- Check Intune device compliance status.
2. Examine wireless logs
- On Windows: netsh wlan show hostednetwork; netsh wlan show interfaces; Review Event Viewer -> Microsoft-Windows-WLAN-AutoConfig for errors such as 0x80070002 (certificate not found) or 0x80090016 (certificate expired).
- On macOS: Console.app -> system.log -> search for "airportd" and "8021X".
3. Validate machine certificate
- Ensure the certificate is present, not expired, and chains to the trusted root CA.
- If missing or expired, initiate a certificate renewal via Intune or instruct the user to run the self-service certificate renewal portal.
4. Check adapter power-saving and driver version
- Confirm power-saving is disabled (as per self-help step).
- Compare driver version against the approved baseline in the Driver Repository; update if outdated.
5. Test 802.1X authentication manually
- Use the command `netsh wlan connect name=ACME-Corp` (Windows) or `networksetup -setairportnetwork en0 ACME-Corp` (macOS) and capture any authentication failure messages.
- If authentication fails, verify that the RADIUS server (NPS) is reachable and that the machine certificate is presented correctly.
6. Assess RF environment
- Use a site-survey tool (e.g., Ekahau Sidekick) or the built-in wireless diagnostics to check channel utilization, interference, and signal strength in the reported location.
- If RSSI <-70?dBm or high co-channel interference, consider adjusting AP tx power or adding an AP.
7. Guest-specific checks (if applicable)
- Verify that the sponsor voucher is active and not expired.
- Ensure the guest portal radius server is responding and that the device receives an IP via DHCP.
8. Resolution and verification
- Once the underlying cause is addressed, ask the user to reconnect and confirm stable connectivity for at least 15?minutes.

- Close the ticket with a summary of the fix and any preventive recommendations (e.g., schedule driver updates, monitor certificate expiry).

Verification

After completing either the self-help or service-desk procedure, perform the following checks to ensure the wireless connection is reliable:

- Ping the internal gateway (e.g., ping 10.0.0.1) and an external host (e.g., ping 8.8.8.8) - both should return <30?ms loss-free replies.
- Browse to the IT portal (https://it.acme.example) and to an external site (https://www.microsoft.com) to validate both intranet and Internet access.
- Run a short speed test (e.g., fast.com) - expect >10?Mbps download in typical office areas; lower speeds may indicate RF issues requiring further investigation.
- Observe the connection for at least 10-15?minutes; note any drops or reconnection events in the wireless logs.

Additional Tips and Best Practices

To minimize future Wi-Fi problems, encourage users to:

- Keep their laptops plugged in or disable aggressive power-saving when using Wi-Fi for extended periods (e.g., in meetings).
- Periodically check the expiration date of their machine certificate (Intune -> Devices -> Certificates) and renew before expiry.
- Report persistent slowness or drops promptly so Network Support can adjust AP configurations or add capacity.
- Use the wired Ethernet dock when available for bandwidth-intensive tasks (video conferencing, large file transfers).

Guest users should always obtain a fresh sponsor voucher before visiting; expired vouchers cause authentication failures even if the SSID is visible.

When to escalate

Escalate to the Network Support group when:

- The user has completed the self-help procedure and still experiences any of the following: repeated credential prompts, no Internet after successful association, frequent disconnections (>2 per hour), or sustained low throughput (<2?Mbps) in areas where other users have normal service.
- Logs indicate certificate validation failures, RADIUS timeout, or authentication errors that cannot be resolved by renewing the certificate locally.
- The issue appears to be localized to a specific AP or area, suggesting a potential RF or hardware problem requiring site survey or AP replacement.

When escalating, include in the ticket:

- Device hostname, asset tag, OS version, and wireless adapter model.
- Screenshot or text of any error messages (e.g., "Windows cannot connect to ACME-Corp", "Authentication failed: 0x80090016").
- Output of `netsh wlan show interfaces` (Windows) or `networksetup -getairportnetwork en0` (macOS).
- Certificate details (subject, issuer, expiry date) from the local certificate store.
- A brief description of the troubleshooting steps already performed and their results.

Keywords: Wi-Fi, 802.1X, ACME-Corp, ACME-Guest, machine certificate, certificate renewal, wireless driver, power saving, forget network, rejoin Wi-Fi, no Internet, limited connectivity, guest access, sponsor voucher, RADIUS, NPS, Intune, Network Support, Service Desk, troubleshooting